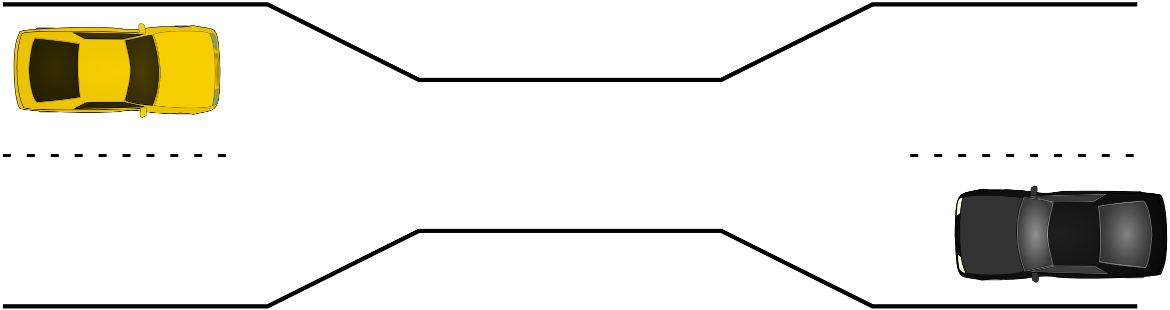
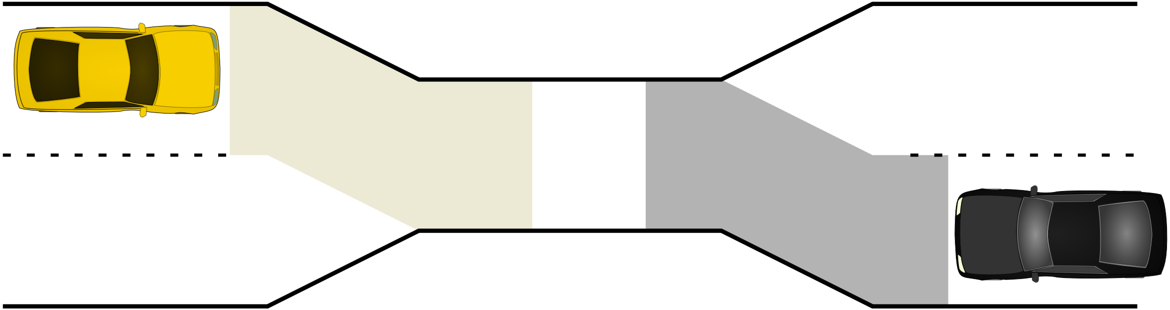
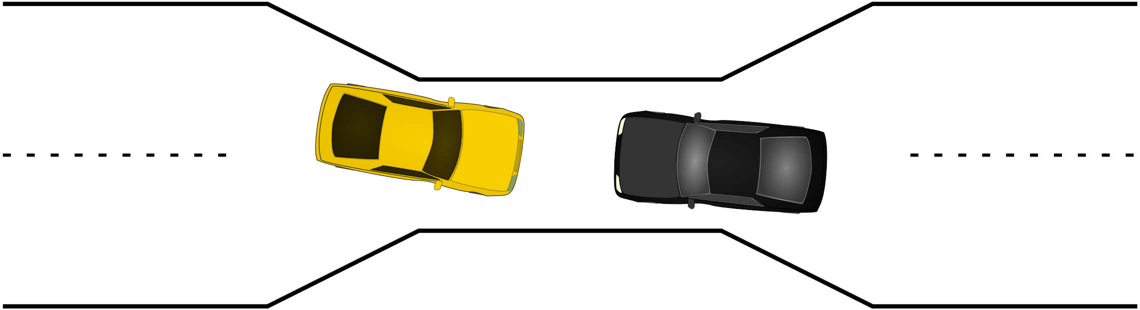
the one-way bridge



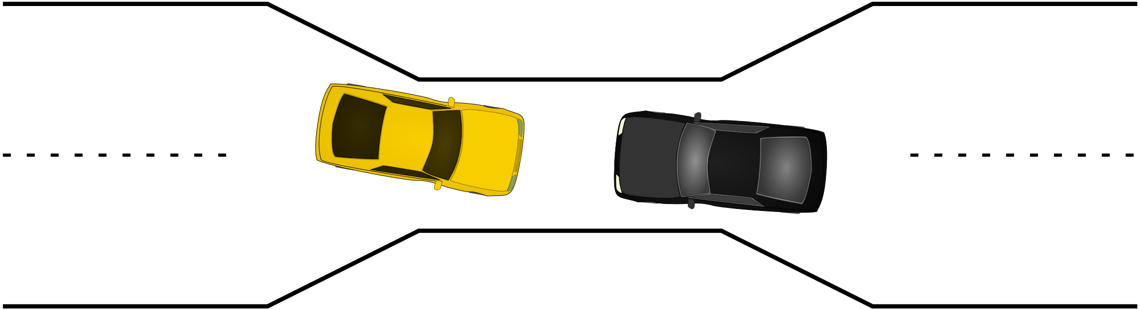
the one-way bridge



the one-way bridge



the one-way bridge



moving two files

```
struct Dir {
  mutex_t lock; HashMap entries;
};
void MoveFile(Dir *from_dir, Dir *to_dir, string filename) {
  mutex_lock(&from_dir->lock);
  mutex_lock(&to_dir->lock);

  Map_put(to_dir->entries, filename,
        Map_get(from_dir->entries, filename));
  Map_erase(from_dir->entries, filename);

  mutex_unlock(&to_dir->lock);
  mutex_unlock(&from_dir->lock);
}
```

Thread 1: `MoveFile(A, B, "foo")`

Thread 2: `MoveFile(B, A, "bar")`

moving two files: lucky timeline (1)

Thread 1 `MoveFile(A, B, "foo")`	Thread 2 `MoveFile(B, A, "bar")`
`lock(&A->lock);`	
`lock(&B->lock);`	
(do move)	
`unlock(&B->lock);`	
`unlock(&A->lock);`	
	`lock(&B->lock);`
	`lock(&A->lock);`
	(do move)
	`unlock(&B->lock);`
	`unlock(&A->lock);`

moving two files: lucky timeline (2)

Thread 1 `MoveFile(A, B, "foo")`	Thread 2 `MoveFile(B, A, "bar")`
`lock(&A->lock);`	
`lock(&B->lock);`	
	`lock(&B->lock…`
(do move)	(waiting for B lock)
`unlock(&B->lock);`	
	`lock(&B->lock);`
	`lock(&A->lock…`
`unlock(&A->lock);`	
	`lock(&A->lock);`
	(do move)
	`unlock(&A->lock);`

moving two files: unlucky timeline

Thread 1	Thread 2
`MoveFile(A, B, "foo")`	`MoveFile(B, A, "bar")`
`lock(&A->lock);`	
	`lock(&B->lock);`

moving two files: unlucky timeline

Thread 1	**Thread 2**
`MoveFile(A, B, "foo")`	`MoveFile(B, A, "bar")`
`lock(&A->lock);`	
	`lock(&B->lock);`
`lock(&B->lock…` *stalled*	
(waiting for lock on B)	`lock(&A->lock…` *stalled*
(waiting for lock on B)	(waiting for lock on A)

moving two files: unlucky timeline

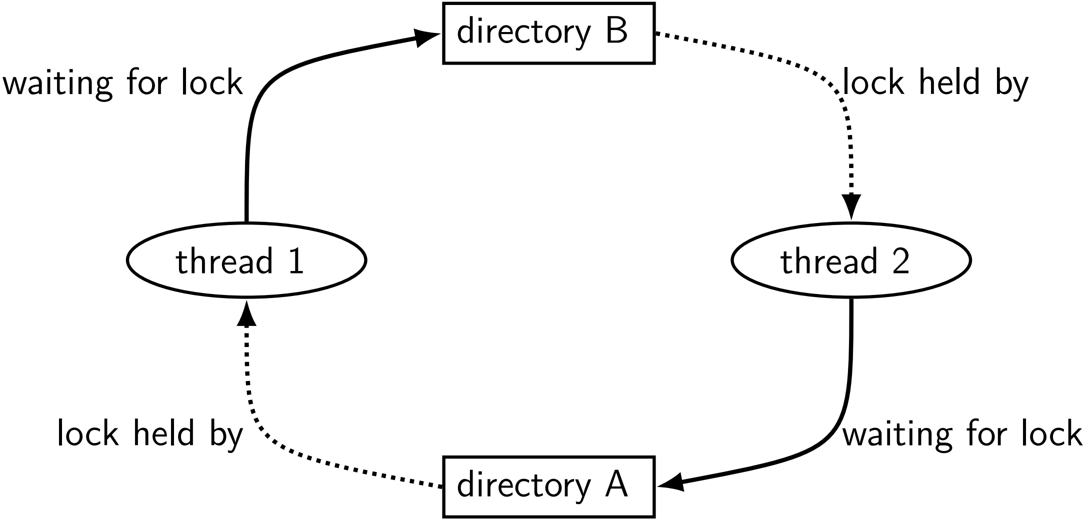
Thread 1 `MoveFile(A, B, "foo")`	**Thread 2** `MoveFile(B, A, "bar")`
`lock(&A->lock);`	
	`lock(&B->lock);`
`lock(&B->lock`… *stalled*	
(waiting for lock on B)	`lock(&A->lock`… *stalled*
(waiting for lock on B)	(waiting for lock on A)
~~(do move)~~ *unreachable*	~~(do move)~~ *unreachable*
~~`unlock(&B->lock);`~~ *unreachable*	~~`unlock(&A->lock);`~~ *unreachable*
~~`unlock(&A->lock);`~~ *unreachable*	~~`unlock(&B->lock);`~~ *unreachable*

moving two files: unlucky timeline

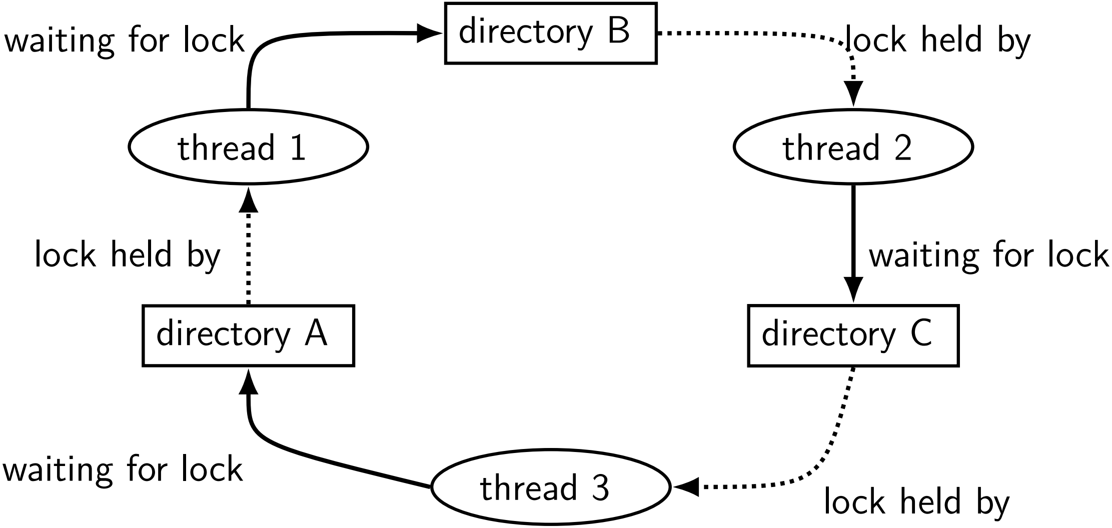
Thread 1	**Thread 2**
`MoveFile(A, B, "foo")`	`MoveFile(B, A, "bar")`
`lock(&A->lock);`	
	`lock(&B->lock);`
`lock(&B->lock…` *stalled*	
(waiting for lock on B)	`lock(&A->lock…` *stalled*
(waiting for lock on B)	(waiting for lock on A)
~~(do move)~~ *unreachable*	~~(do move)~~ *unreachable*
~~`unlock(&B->lock);`~~ *unreachable*	~~`unlock(&A->lock);`~~ *unreachable*
~~`unlock(&A->lock);`~~ *unreachable*	~~`unlock(&B->lock);`~~ *unreachable*

Thread 1 holds A lock, waiting for Thread 2 to release B lock

moving two files: dependencies



moving three files: dependencies



moving three files: unlucky timeline

Thread 1	**Thread 2**	**Thread 3**
`MoveFile(A, B, "foo")`	`MoveFile(B, C, "bar")`	`MoveFile(C, A, "quux")`
`lock(&A->lock);`		
	`lock(&B->lock);`	
		`lock(&C->lock);`
`lock(&B->lock…` *stalled*		
	`lock(&C->lock…` *stalled*	
		`lock(&A->lock…` *stalled*

deadlock with free space

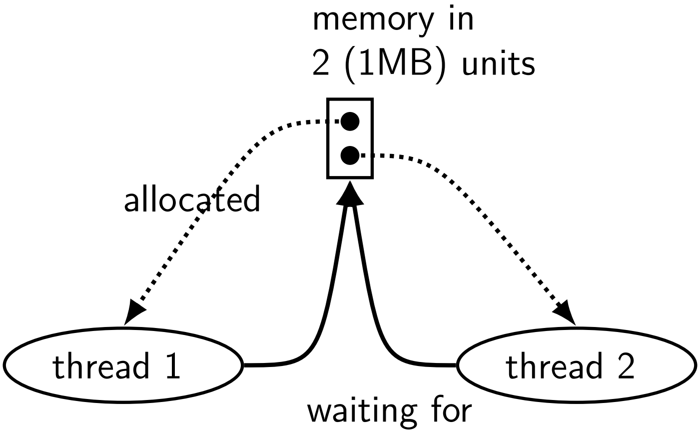
Thread 1	**Thread 2**
`AllocateOrWaitFor(1 MB)`	`AllocateOrWaitFor(1 MB)`
`AllocateOrWaitFor(1 MB)`	`AllocateOrWaitFor(1 MB)`
`(do calculation)`	`(do calculation)`
`Free(1 MB)`	`Free(1 MB)`
`Free(1 MB)`	`Free(1 MB)`

2 MB of space — deadlock possible with unlucky order

deadlock with free space (unlucky case)

Thread 1	Thread 2
`AllocateOrWaitFor(1 MB)`	
	`AllocateOrWaitFor(1 MB)`
`AllocateOrWaitFor(1 MB…` *stalled*	
	`AllocateOrWaitFor(1 MB…` *stalled*

free space: dependency graph



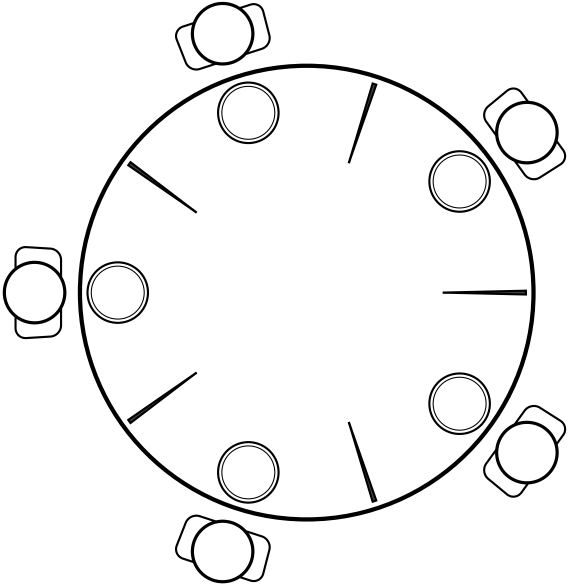
deadlock with free space (lucky case)

Thread 1	Thread 2
`AllocateOrWaitFor(1 MB)`	
`AllocateOrWaitFor(1 MB)`	
`(do calculation)`	
`Free(1 MB);`	
`Free(1 MB);`	
	`AllocateOrWaitFor(1 MB)`
	`AllocateOrWaitFor(1 MB)`
	`(do calculation)`
	`Free(1 MB);`
	`Free(1 MB);`

lab next week

applying solutions to deadlock to classic *dining philosphers* problem

dining philosophers



five philosophers either think or eat

to eat:

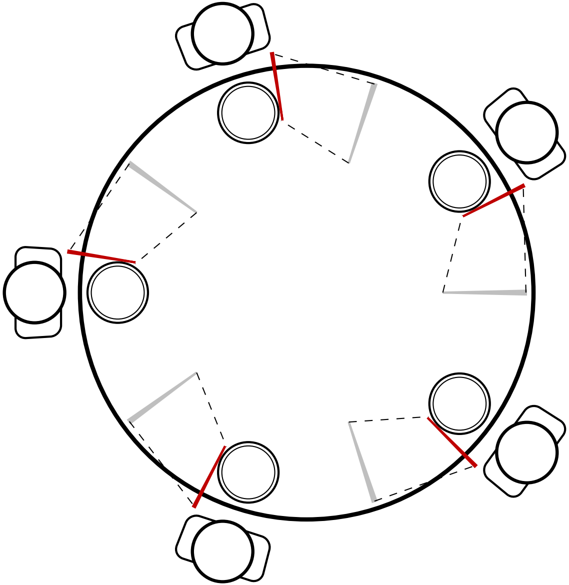
grab chopstick on left, then

grab chopstick on right, then

then eat, then

return chopsticks

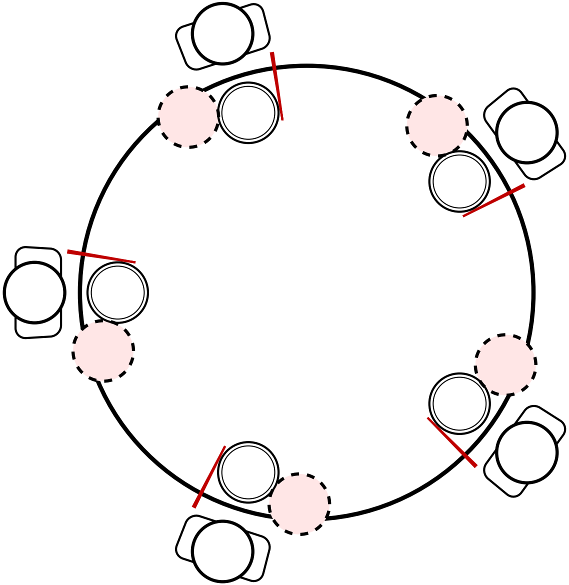
dining philosophers



everyone eats at the same time?

grab left chopstick, then…

dining philosophers



everyone eats at the same time?
grab left chopstick, then
try to grab right chopstick, ...
we're at an impasse

deadlock

deadlock — circular waiting for resources

resource = something needed by a thread to do work

- locks
- CPU time
- disk space
- memory
- ...

often non-deterministic in practice

most common example: *when acquiring multiple locks*

deadlock

deadlock — circular waiting for *resources*

resource = something needed by a thread to do work

- locks
- CPU time
- disk space
- memory
- ...

often non-deterministic in practice

most common example: *when acquiring multiple locks*

deadlock requirements

mutual exclusion

one thread at a time can use a resource

hold and wait

thread holding a resources waits to acquire *another* resource

no preemption of resources

resources are only released voluntarily

thread trying to acquire resources can't 'steal'

circular wait

there exists a set $\{T_1, \ldots, T_n\}$ of waiting threads such that

- T_1 is waiting for a resource held by T_2
- T_2 is waiting for a resource held by T_3
- …
- T_n is waiting for a resource held by T_1

how is deadlock possible?

Given list: A, B, C, D, E

```
RemoveNode(LinkedListNode *node) {
    pthread_mutex_lock(&node->lock);
    pthread_mutex_lock(&node->prev->lock);
    pthread_mutex_lock(&node->next->lock);
    node->next->prev = node->prev; node->prev->next = node->next;
    pthread_mutex_unlock(&node->next->lock); pthread_mutex_unlock(&node->p
    pthread_mutex_unlock(&node->lock);
}
```

Which of these (all run in parallel) can deadlock?

A. RemoveNode(B) and RemoveNode(C)
B. RemoveNode(B) and RemoveNode(D)
C. RemoveNode(B) and RemoveNode(C) and RemoveNode(D)
D. A and C
E. B and C
F. all of the above
G. none of the above

how is deadlock — solution

Remove B	Remove C
lock B	lock C
lock A (prev)	wait to lock B (prev)
wait to lock C (next)	

With B and D — only overlap in in node C — no circular wait possible
(thread can't be waiting while holding something other thread wants)

deadlock prevention techniques

infinite resources or at least enough that never run out	no *mutual exclusion*
no shared resources	no *mutual exclusion*
no waiting "busy signal" — abort and (maybe) retry revoke/preempt resources	no *hold and wait*/ *preemption*
acquire resources in **consistent order**	no *circular wait*
request **all resources at once**	no *hold and wait*

deadlock prevention techniques

infinite resources or at least enough that never run out	no *mutual exclusion*
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acquire resources in **consistent order**	no *circular wait*
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deadlock prevention techniques

infinite resources

> memory allocation: malloc() fails rather than waiting (no deadlock)
> locks: `pthread_mutex_trylock` fails rather than waiting
> problem: retry how many times? *no bound on number of tries needed*
> ...

exclusion

exclusion

no waiting

"*busy signal*" — *abort and (maybe) retry*

revoke/preempt resources

no *hold and wait*/*preemption*

acquire resources in **consistent order** — no *circular wait*

request **all resources at once** — no *hold and wait*

abort and retry limits?

abort-and-retry

pthread's mutexes:

```
pthread_mutex_trylock
pthread_mutex_timedlock
```

how many times will you retry?

moving two files: abort-and-retry

```
struct Dir { mutex_t lock; HashMap entries; };
void MoveFile(Dir *from_dir, Dir *to_dir, string filename) {
  while (true) {
    mutex_lock(&from_dir->lock);
    if (mutex_trylock(&to_dir->lock) == LOCKED) break;
    mutex_unlock(&from_dir->lock);
  }

  Map_put(to_dir->entries, filename, Map_get(from_dir->entries, fil
  from_dir->entries.erase(filename);

  mutex_unlock(&to_dir->lock);
  mutex_unlock(&from_dir->lock);
}
```

Thread 1: `MoveFile(A, B, "foo")`; Thread 2: `MoveFile(B, A, "bar")`

moving two files: lots of bad luck?

Thread 1 `MoveFile(A, B, "foo")`	Thread 2 `MoveFile(B, A, "bar")`
`lock(&A->lock)` → `LOCKED`	
	`lock(&B->lock)` → `LOCKED`
`trylock(&B->lock)` → `FAILED`	
	`trylock(&A->lock)` → `FAILED`
`unlock(&A->lock)`	
	`unlock(&B->lock)`
`lock(&A->lock)` → `LOCKED`	
	`lock(&B->lock)` → `LOCKED`
`trylock(&B->lock)` → `FAILED`	
	`trylock(&A->lock)` → `FAILED`
`unlock(&A->lock)`	

livelock

livelock: keep aborting and retrying without end

like deadlock — no one's making progress

potentially forever

unlike deadlock — threads are not waiting

deadlock prevention techniques

infinite resources or at least enough that never run out	no *mutual exclusion*
no shared resources	no *mutual exclusion*
no waiting "*busy signal*" — *abort and (maybe) retry* revoke/preempt resources	no *hold and wait*/ *preemption*
acquire resources in **consistent order**	no *circular wait*
request **all resources at once**	no *hold and wait*

preventing livelock

make schedule random — e.g. random waiting after abort

make threads run one-at-a-time if lots of aborting

other ideas?

deadlock prevention techniques

infinite resources — no *mutual exclusion*

or at least enough that never run out

no share[…] — […]*exclusion*

requires some way to undo partial changes to avoid errors

common approach for databases

...

no waiti[…] — no *hold and wait*/*preemption*

"busy signal" — abort and (maybe) retry

revoke/preempt resources

acquire resources in **consistent order** — no *circular wait*

request **all resources at once** — no *hold and wait*

stealing locks???

how do we make stealing locks possible

unclean: just kill the thread

- problem: inconsistent state?

clean: have code to undo partial oepration

- some databases do this

won't go into detail in this class

revokable locks?

```
try {
    AcquireLock();
    use shared data
} catch (LockRevokedException le) {
    undo operation hopefully?
} finally {
    ReleaseLock();
}
```

deadlock prevention techniques

infinite resources — no *mutual exclusion*

or at least enough that never run out

no shared resources — no *mutual exclusion*

no waiting — no *hold and wait*/ *preemption*

"busy signal" — abort and (maybe) retry

revoke/preempt resources

acquire resources in ***consistent order*** — no *circular wait*

request **all resources at once** — no *hold and wait*

acquiring locks in consistent order (1)

```
MoveFile(Dir* from_dir, Dir* to_dir, string filename) {
  if (from_dir->path < to_dir->path) {
    lock(&from_dir->lock);
    lock(&to_dir->lock);
  } else {
    lock(&to_dir->lock);
    lock(&from_dir->lock);
  }
  ...
}
```

acquiring locks in consistent order (1)

```
MoveFile(Dir* from_dir, Dir* to_dir, string filename) {
  if (from_dir->path < to_dir->path) {
    lock(&from_dir->lock);
    lock(&to_dir->lock);
  } else {
    lock(&to_dir->lock);
    lock(&from_dir->lock);
  }
  ...
}
```

any ordering will do
e.g. compare pointers

acquiring locks in consistent order (2)

often by convention, e.g. Linux kernel comments:

```
/*
 * ...
 * Lock order:
 *      contex.ldt_usr_sem
 *        mmap_sem
 *          context.lock
 */
```

```
/*
 * ...
 * Lock order:
 *   1. slab_mutex (Global Mutex)
 *   2. node->list_lock
 *   3. slab_lock(page) (Only on some arches and for debugging)
 * ...
 */
```

deadlock prevention techniques

infinite resources or at least enough that never run out	no *mutual exclusion*
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